

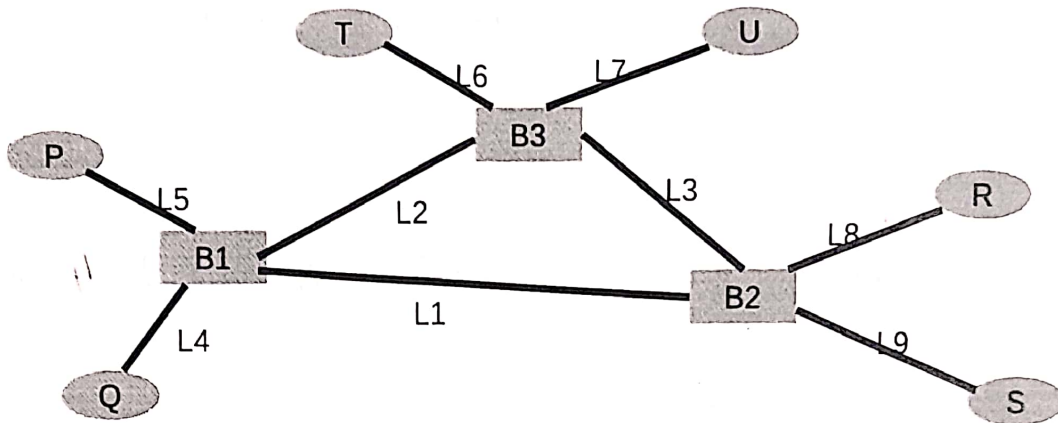
CS348 Computer Networks Quiz-2, Thu 10 Sep 2026, 09.30am-10.10am, Max. Marks: 10

Name: SAMPLE ANSWERS Roll number: br12345 House: 1

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only. You do NOT need a calculator.
- All answers have to be (briefly) explained. State any necessary assumptions.
- The final answer of each question has to be entered in the given box.

Q1: [1 x 4 = 4 marks] Learning bridges and spanning tree: In the ethernet network below, B1, B2, B3 are bridges and other nodes P, Q, R, S, T, U are end-hosts. The various links are labeled as shown. In the questions below, assume: (a) spanning tree protocol has stabilized, (b) forwarding tables are empty to begin with.



1. Which port of which bridge is inactive (if any)?

Answer: B3-L3

Reason: All ports toward hosts are DP, active. B2-L3 = DP
All ports of B1 = RP, active. B2-L1, B3-L2 = RP, B3-L3 = IN

2. Node P transmits a frame destined to node S. How many links is the frame sent on (assume no losses)?

Answer: 9

Reason: All links including B2-L3 (sent by B2)

3. After above, S transmits a frame back to P. How many links is this frame sent on (assume no losses)?

Answer: 3

Reason: B1 & B2 have learnt path to P. So: L9, L1, L5

4. After above, T transmits a frame to S. How many links is this frame sent on (assume no losses)?

Answer: 5

Reason: B1, B2 have learnt path to S, but not B3
So L6, L2, L1, L9 and L7

Q2: IP fragmentation and pipelining [1+1+2+1+1=6 marks]: In the network below, N1 sends a packet P1 with a 620 byte TCP payload to N2 via IP router R. The two links N1-R and R-N2 are both of the same technology, with a link capacity of 4Mbps. Assume that TCP and IP headers are both 20 bytes each. Ignore propagation and processing delays, and assume that there is no other traffic in the network.



1. What is the minimum required MTU of the link technology so that P1 does not get fragmented by IP?

Answer:

660

Reason: $IP \text{ payload} = 620 + 20$

$Link \text{ payload} = 620 + 20 + 20 = 660 \text{ bytes}$

$MTU \geq 660 \text{ for no frag}$

2. What is the delay (in ms) experienced by P1, from N1 to N2, in the above no-fragmentation scenario?

Answer:

2.640

Calculation: $Only \text{ Tx delay} = \frac{660 \times 8}{4 \text{ m}} \times 2 \text{ links} = 2.640 \text{ ms}$

3. For what value of MTU of the link technology does P1 get fragmented into two IP fragments of same size?

Answer:

340

Reason: $IP \text{ payload} = 640 \rightarrow 2 \text{ frags} \Rightarrow 320 \text{ bytes each}$

Note: 320 is a multiple of 8

$MTU \text{ has to be } 320 + 20 = 340 \text{ bytes}$

for such a split to happen

4. What is the delay (in ms) experienced by P1, from N1 to N2, in the above two-fragments scenario?

Answer:

2.040

Calculation: Note: Tx of frag1 on R-N2 is in parallel

with Tx of frag2 on N1-R. So: $\frac{340 \times 8}{4 \text{ m}} \times 3 = 2.040 \text{ ms}$

5. In which of the two scenarios (no-fragmentation vs two-fragments) is P1's delay lower? Explain the intuition behind the same.

Answer:

Two-fragments

Explanation: Pipelining effect: above mentioned parallelism results in lower latency despite higher header overhead